Goal: Solve optimization problems involving volume, surface area, and distances.

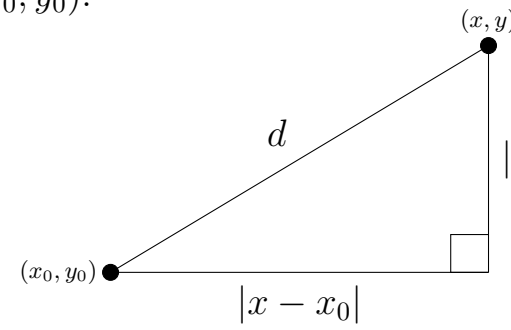
Example 1. We want to construct a box with a square base and no top. If the volume is 500 cubic inches, then what is the minimum amount of material required?

Example 2 (Ideal Soup Can)**.** A company is designing a cylindrical soup can to hold $250\pi \approx 785.4\text{cm}^3$ of liquid. What should the dimensions be to use the least amount of material?

Example 3 (Ideal Soup Can II)**.** Due to the manufacturing process, each can has to use $294\pi \approx 923.63\text{cm}^2$ of aluminum. Find the largest amount of soup these cans could store.

Distances

Recall the formula for computing the distance between the points (x, y) and (x_0, y_0):



$$d = \sqrt{(x - x_0)^2 + (y - y_0)^2}$$

Since distances are positive, squaring and taking square roots preserve inequality. I.e., if there is a minimum distance:

$$\begin{aligned} [\text{minimum}]^2 &\leq [d]^2 \\ \sqrt{[\text{minimum}]^2} &\leq \sqrt{[d]^2} \\ \text{minimum} &\leq d \end{aligned}$$

In short: Finding the location where the optimal distance occurs is the same as finding the location where the _____ occurs.

Example 4. What is the minimum distance between $y = 7 - 2x$ and the point $(1, 0)$?

Example 5. Where is the distance between $(0, \frac{1}{2})$ and the parabola $y = x^2 - 1$ at a minimum?